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United States Patent Application Publication

20250285859

Kind Code

A1

Publication Date

September 11, 2025

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METHOD FOR MANUFACTURING GROUP 3 NITRIDE SEMICONDUCTOR TEMPLATE

Abstract

The present invention relates to a method for manufacturing a group 3 nitride semiconductor template, the method comprising: a growing step of growing a seed layer with a nitrogen polar surface as an upper surface on a growth substrate; a depositing step of depositing a protective layer on the seed layer; a bonding step of bonding the protective layer and a support substrate through a bonding layer; and a removing step of removing the growing substrate to expose a metal polar surface of the seed layer.

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Appl. No.: 18/860654

Filed (or PCT Filed): November 02, 2023

PCT No.: PCT/KR2023/017365

Foreign Application Priority Data

KR 10-2022-0148587

Nov. 09, 2022

Publication Classification

Int. Cl.: H01L21/02 (20060101); C30B25/18 (20060101); C30B29/40 (20060101); C30B33/00 (20060101)

U.S. Cl.:

CPC **H01L21/0254** (20130101); **C30B25/183** (20130101); **C30B25/186** (20130101);
C30B29/403 (20130101); **C30B33/00** (20130101); **H01L21/02381** (20130101);
H01L21/0242 (20130101); **H01L21/02488** (20130101);

Background/Summary

TECHNICAL FIELD

[0001] The present invention relates to a method of manufacturing a group III nitride semiconductor template.

BACKGROUND ART

[0002] Unlike SiC semiconductor devices that are grown, designed, and manufactured on the conventional homogeneous single-crystalline SiC growth substrate (wafer), group III nitride semiconductor devices are designed and manufactured by epitaxially growing a group III nitride semiconductor directly on a heterogeneous large-diameter (6 inches or larger) single-crystalline sapphire, SiC, or Si growth substrate due to the absence of low-cost commercial single-crystalline GaN and AlN growth substrates, and due to a difference in lattice constant (LC) with the growth substrate, various high-density crystallographic defects occur, which degrades the performance and quality of the semiconductor device, making it very difficult to expand the scope of application.

[0003] Specifically, in a technique of epitaxially growing a group III nitride semiconductor directly on a heterogeneous large-diameter single-crystalline sapphire, SiC, or Si growth substrate, three factors such as a surface temperature difference (ΔT) between upper and lower portions of the growth substrate, an LC difference (Δa), and a coefficient of thermal expansion difference ($\Delta \alpha$) induce stress in the group III nitride epitaxy in a stepwise manner, resulting in crystallographic defects, concave or convex bowing, etc.

[0004] Due to the above-described three stress influencing factors, the following problems occur when group III nitride material-based device products are manufactured.

[0005] First, when micro light emitting diode (LED) devices are grown, the epitaxy wafer bowing phenomenon occurs due to the above-described three stress influencing factors, and thus a surface temperature difference between the center and edge regions causes non-uniformity of an indium (In) composition ratio during the growth of an InGaN-based active layer (multi quantum wells (MQWs)), which greatly disperses the wavelength and photoelectric characteristics (operating voltage, optical output) in the wafer and significantly affects the yield of good products, resulting in increased manufacturing costs. In addition, the wafer bowing phenomenon lowers the uniformity of the aluminum (Al) composition ratio and the doping amount of magnesium (Mg) that is a p-type dopant atom in the p-type AlGaN that serves as an electron blocking layer that is continuously subsequently grown during the growth of micro LED devices with the InGaN-based active layer (MQWs), resulting in the issue of dispersion of the photoelectric characteristics in the wafer.

[0006] In addition, even when a power semiconductor device is grown, the above three stress influencing factors cause the epitaxy wafer bowing phenomenon, and thus there is a problem that the uniformity of the thickness and aluminum (Al) composition ratio of the AlGaN barrier with a thickness of about 20 nm in a high electron mobility transistor (HEMT) with a horizontal channel structure is degraded, tensile stress further becomes severe during the growth of a thick GaN of 10 μm or more for a power semiconductor device with a vertical drift structure, thereby causing quality degradation, and wafer bowing becomes severe during wafer cooling to room temperature after growth, thereby increasing the possibility of cracks.

[0007] In addition, even during the growth of a communication filter device, such as bulk acoustic

wave (BAW) or surface acoustic wave (SAW) filters, made of AlN materials, in a filter device in which the crystallinity and thickness uniformity of AlN with high piezoelectricity significantly affect quality, there is a problem that the above three stress influencing factors cause the epitaxy wafer bowing phenomenon, and there is a problem that many cracks and quality degradation occur inside AlN around 500 nm due to strong tensile stress during the growth of AlN.

DISCLOSURE

Technical Problem

[0008] The present invention is intended to solve the above-described conventional problems and is directed to providing a method of manufacturing a group III nitride semiconductor template, which may manufacture a high-quality group III nitride semiconductor device by growing a group III nitride semiconductor seed layer having a nitrogen polar surface on an initial growth substrate and transferring the group III nitride semiconductor seed layer onto a final support substrate having high heat resistance and high corrosion resistance to have an indium (In), gallium (Ga), or aluminum (Al) metal polar surface.

Technical Solution

[0009] According to the present invention, the above object is achieved by a method of manufacturing a group III nitride semiconductor template, which includes a growing operation of growing a seed layer whose upper surface is a nitrogen polar surface on a growth substrate, a depositing operation of depositing a protective layer on the seed layer, a bonding operation of bonding the protective layer to a support substrate through a bonding layer, and a removing operation of removing the growth substrate to expose a metal polar surface of the seed layer.

[0010] In addition, the growth substrate may be a silicon (Si) substrate.

[0011] In addition, the growing operation may include performing high-temperature heat treatment on the growth substrate in a hydrogen atmosphere, performing high-temperature nitriding treatment on a surface of the growth substrate in an ammonia atmosphere to form a SiN layer, depositing a buffer layer on the SiN layer, growing a compressive stress introduction layer on the buffer layer, and growing a seed layer having a nitrogen polar surface on the compressive stress introduction layer.

[0012] In addition, the removing operation may include removing the growth substrate, the SiN layer, the buffer layer, and the compressive stress introduction layer to expose the metal polar surface of the seed layer.

[0013] In addition, Mg doping or carrier gas modulation may be performed on at least one of the compressive stress introduction layer and the seed layer during growth.

[0014] In addition, the removing operation may include removing the growth substrate, the SiN layer, the buffer layer, the compressive stress introduction layer, and a region of the seed layer on which the Mg doping or carrier gas modulation is performed to expose the metal polar surface of the seed layer.

[0015] In addition, the growth substrate may be a sapphire (Al₂O₃) substrate.

[0016] In addition, the growing operation may include a first operation of performing high-temperature heat treatment on the growth substrate in a hydrogen atmosphere, a second operation of performing high-temperature nitriding treatment on a surface of the growth substrate in an ammonia atmosphere to form an AlN layer, a third operation of depositing a buffer layer on the AlN layer, and a fourth operation of growing a seed layer having a nitrogen polar surface on the buffer layer.

[0017] In addition, the removing operation may include removing the growth substrate, the AlN layer, and the buffer layer to expose the metal polar surface of the seed layer.

[0018] In addition, Mg doping or carrier gas modulation may be performed on the seed layer during growth.

[0019] In addition, the removing operation may include removing the growth substrate, the AlN layer, the buffer layer, and a region of the seed layer on which the Mg doping or carrier gas

modulation is performed to expose the metal polar surface of the seed layer.

[0020] In addition, the depositing operation may include forming a reinforcement layer, which increases bonding strength and induces compressive stress, on the seed layer and then depositing the protective layer on the reinforcement layer.

[0021] In addition, the bonding operation may include forming a reinforcement layer, which increases bonding strength and induces compressive stress, on the support substrate and then bonding the protective layer to the reinforcement layer through a bonding layer.

[0022] In addition, the method according to the present invention may further include a re-growing operation of re-growing an device active layer on the exposed seed layer.

Advantageous Effects

[0023] According to the present invention, since a homogeneous seed layer without a lattice constant difference can be formed on a final support substrate having high heat resistance and high corrosion resistance through dielectric bonding and epitaxy transfer processes, it is possible to significantly improve the performance and quality of a group III nitride semiconductor device grown on the seed layer.

[0024] In addition, according to the present invention, since the method can use not only single crystalline but also a polycrystalline microstructure resulting from a high-temperature sintering process and can also be applied to a wafer having a size of 8 inches or more, it is possible to secure cost competitiveness.

[0025] Meanwhile, the effects of the present invention are not limited to the above-described effects, and may include various effects within a range that is apparent to those skilled in the art from the following descriptions.

Description

DESCRIPTION OF DRAWINGS

[0026] FIG. 1 is a flowchart of a method of manufacturing a group III nitride semiconductor template according to a first embodiment of the present invention.

[0027] FIG. 2 shows a process of manufacturing a group III nitride semiconductor template according to the first embodiment of the present invention.

[0028] FIGS. 3 and 4 show that a seed layer whose upper surface is a nitrogen polar surface is grown through a growing operation of the method of manufacturing a group III nitride semiconductor template according to the first embodiment of the present invention.

[0029] FIG. 5 is a flowchart of a method of manufacturing a group III nitride semiconductor template according to a second embodiment of the present invention.

[0030] FIG. 6 shows a process of manufacturing a group III nitride semiconductor template according to the second embodiment of the present invention.

[0031] FIGS. 7 and 8 show that a seed layer whose upper surface is a nitrogen polar surface is grown through a growing operation of the method of manufacturing a group III nitride semiconductor template according to the second embodiment of the present invention.

[0032] FIG. 9 shows that an device active layer is re-grown on the seed layer of the group III nitride semiconductor template according to the method of manufacturing a group III nitride semiconductor template according to the first or second embodiment of the present invention.

MODES OF THE INVENTION

[0033] Hereinafter, some embodiments of the present invention will be described in detail with reference to exemplary drawings. In adding reference numerals to components in each drawing, it should be noted that the same components have the same reference numerals as much as possible even when they are illustrated in different drawings.

[0034] In addition, in describing embodiments of the present invention, detailed descriptions of

related known configurations or functions will be omitted when it is determined that the detailed descriptions obscure the understanding of the embodiments of the present invention.

[0035] In addition, terms such as first, second, A, B, (a), and (b) may be used to describe components of the embodiments of the present invention. These terms are only for the purpose of distinguishing one component from another component, and the nature, sequence, order, or the like of the corresponding component is not limited by the terms.

[0036] Group III nitride semiconductor devices are designed and manufactured by epitaxially growing a group III nitride semiconductor directly on a heterogeneous single-crystalline substrate wafer due to the absence of low-cost commercial single-crystalline GaN and AlN growth substrates, and due to a difference in lattice constant (LC) with an initial growth substrate, various high-density crystallographic defects occur, which degrades the performance and quality of the semiconductor device, making it very difficult to expand the scope of application.

[0037] The present invention is intended to solve such a problem and to minimize defects when a group III nitride semiconductor is grown by forming a homogeneous growth seed layer having an LC difference of zero on a final support substrate having high heat resistance and high corrosion resistance through dielectric wafer bonding and epitaxy transfer processes, thereby significantly improving the performance and quality of a group III nitride semiconductor device. However, when the growth seed layer is transferred onto the final support substrate, an exposed surface of the growth seed layer has a nitrogen polar surface, making it very difficult to grow another high-quality group III nitride semiconductor thereon.

[0038] Therefore, the present invention is characterized in that the growth seed layer is first grown to have a nitrogen polar surface on the initial growth substrate through processes of controlling crystallographic face and orientation angle, performing high-temperature surface nitriding treatment before growth, growing a buffer layer or nucleation layer, etc., and then the surface of the growth seed layer transferred onto the final support substrate has a metal (Ga, Al, In) polar surface through the dielectric bonding and epitaxy transfer processes.

[0039] A method **S100** of manufacturing a group III nitride semiconductor template according to the first embodiment of the present invention will now be described in detail with reference to the accompanying drawings.

[0040] FIG. 1 is a flowchart of a method **S100** of manufacturing a group III nitride semiconductor template according to the first embodiment of the present invention, and FIG. 2 shows a process of manufacturing a group III nitride semiconductor template according to the first embodiment of the present invention.

[0041] As shown in FIGS. 1 and 2, the method **S100** of manufacturing a group III nitride semiconductor template according to the first embodiment of the present invention includes a growing operation **S110**, a depositing operation **S120**, a bonding operation **S130**, and a removing operation **S140**.

[0042] The growing operation **S110** is an operation of growing a single-layer or multilayered group III nitride semiconductor seed layer **150** whose upper surface is a nitrogen polar surface on an initial growth substrate **G1**.

[0043] In the present embodiment, the initial growth substrate **G1** may be removed by wet etching during a chemical lift off (CLO) process in the removing operation **S140** to be described below and is provided as an Si growth substrate **G1** that may be mechanically polished and selectively etched, and the Si growth substrate **G1** is preferably made of Si having a (111) crystal face or (100) crystal face so that a high-quality group III nitride semiconductor seed layer **150** may be grown.

[0044] FIGS. 3 and 4 show that the seed layer **150** whose upper surface is a nitrogen polar surface is grown through the growing operation **S110** of the method **S100** of manufacturing a group III nitride semiconductor template according to the first embodiment of the present invention.

[0045] More specifically, the growing operation **S110** includes preparing an initial growth substrate **G1** on which a chemical mechanical polishing (CMP) process is performed, performing heat

treatment on the initial growth substrate **G1** at a high temperature of 1000° C. or higher in a hydrogen atmosphere (H₂), performing nitriding treatment on a surface of the growth substrate **G1** at a high temperature of 1000° C. or higher in an ammonia atmosphere (NH₃ and N₂) to form an SiN layer **P1** having a film or island shape in a thickness of less than 5 nm, depositing an AlN buffer layer **U1** (or a nucleation layer) on the SiN layer **P1** in a thickness of 50 nm or less at a high temperature of 850° C. or higher, growing a Al_xGa_yN (x<y) compressive stress introduction layer **R** for inducing compressive stress on the buffer layer **U1** in a thickness of 500 nm or less at a high temperature of 950° C. or higher, and growing a seed layer **150** to have a thickness of 2 μm or less on the compressive stress introduction layer **R**.

[0046] As shown in FIG. 3, according to the growing operation **S110**, a structure in which the SiN layer **P1**, the AlN buffer layer **U1**, the Al_xGa_yN (x<y) compressive stress introduction layer **R**, and the seed layer **150** are sequentially stacked is formed on the growth substrate **G1**, and through the SiN layer **P1** and the buffer layer **U1** according to high-temperature nitriding treatment, an Al_xGa_yN (x<y) compressive stress introduction layer **R** having a nitrogen polar surface may be grown, and then a seed layer **150** having a nitrogen polar surface may be grown.

[0047] Meanwhile, as shown in FIG. 4, to reinforce the nitrogen polarity of the surface of the seed layer **150**, Mg doping or carrier gas modulation may be performed on at least one of the compressive stress introduction layer **R** and the seed layer **150** during growth, and preferably, Mg doping or carrier gas modulation may be performed on each of the compressive stress introduction layer **R** and the seed layer **150** during growth.

[0048] Meanwhile, a channel through which gas generated during the process may move may be formed inside the seed layer **150** through an intentional patterning design.

[0049] The depositing operation **S120** is an operation of depositing a protective layer **140** on the seed layer **150**.

[0050] Here, the protective layer **140** is a layer for preventing the seed layer **150** from being damaged during a subsequent process, and may contain an oxide including SiO₂ and a nitride including SiN_x, and contain a metal, an alloy, etc.

[0051] Meanwhile, the depositing operation **S120** may include, before depositing the protective layer **140**, forming a reinforcement layer **120** on the seed layer **150** and then depositing the protective layer **140** on the reinforcement layer **120**.

[0052] The bonding operation **S130** is an operation of bonding the protective layer **140** to a support substrate **110** through a bonding layer **130**.

[0053] In the present embodiment, the final support substrate **110** may be made of a material having high heat resistance or high corrosion resistance at a high temperature of 1000° C. or higher and in a reducing atmosphere and specifically, may contain AlN_cera (heat dissipation performance: 170 to 230 W/mK), SiC (heat dissipation performance: 300 to 450 W/mK), Si (heat dissipation performance: 149 W/mK), sapphire (heat dissipation performance: 30 W/mK), and SiN_x (heat dissipation performance: 90 W/mK). That is, in the present invention, not only a single crystalline but also a polycrystalline microstructure resulting from a high-temperature sintering process may be used, thereby securing cost competitiveness.

[0054] The bonding operation **S130** may include bonding the final support substrate **110** to the protective layer **140** through the bonding layer **130** and specifically, may include forming a first bonding layer **B1** on the protective layer **140**, forming a second bonding layer **B2** on the final support substrate **110**, then pressing and bonding the first bonding layer **B1** to the second bonding layer **B2** at 200° C. or lower to form the bonding layer **130**, and bonding the initial growth substrate **G1** on which the protective layer **140** is formed to the final support substrate **110**. In general, the greater the difference in coefficient of thermal expansion between the Si growth substrate **G1** and the final support substrate **110** material, the more preferable it is to decrease the temperature of the bonding operation **S130** to room temperature and increase the pressure, and it is preferable to perform a CMP process on each of the first bonding layer **B1** and the second bonding

layer B2 to increase bonding strength.

[0055] The bonding layer **130** (including the first bonding layer B1 and the second bonding layer B2) may be formed as a single layer or multiple layers, and a dielectric material whose properties do not change and has excellent thermal conductivity in a metal organic chemical vapor deposition (MOCVD) chamber (at a temperature of 1000° C. or higher and in a reducing atmosphere) in which a group III nitride semiconductor is grown is preferentially selected, and for example, SiO₂, SiN_x, SiCN, AlN, Al₂O₃, a-Si (amorphous silicon), and furthermore, a flowable oxide (FOx) such as spin on glass (SOG, liquid SiO₂) or hydrogen silsesquioxane (HSQ) may be selected to decrease surface roughness.

[0056] Meanwhile, a channel through which gas generated during the process may move may be formed inside at least one of the first bonding layer B1 and the second bonding layer B2 through an intentional patterning design.

[0057] When wafer bonding is performed using a dielectric material, the dielectric material is sensitive to the roughness of the wafer bonding surface and wafer bowing, but the above-described patterning inside the bonding layer **130** can significantly decrease the issue of severe wafer surface roughness and bowing. In addition, the above-described patterning inside the bonding layer **130** can effectively prevent the formation of voids inside the bonding layer **130** and can also buffer minor thermal induced stress in addition to increasing bonding strength.

[0058] In the past, epitaxial wafer bowing may occur due to the thermo-mechanical induced stress caused by differences in LC and coefficient of thermal expansion (CTE) between the initial growth substrate G1 and the group III nitride semiconductor, but in the present invention, the above problem can be solved by strongly bonding the final support substrate **110** through the bonding layer **130**. That is, since the epitaxial wafer to which the final support substrate **110** is bonded is in a stress-relieved state and wafer bowing can be minimized to almost zero, there is an advantage that the method can be applied to wafers having sizes of not only 4 inches, 6 inches, and 8 inches, but also 12 inches or more.

[0059] Meanwhile, the bonding operation S130 may include forming the reinforcement layer **120** on the support substrate **110** and then bonding the protective layer **140** to the reinforcement layer **120** through the bonding layer **130**.

[0060] Here, the reinforcement layer **120** is a layer for increasing the bonding strength with the final support substrate **110** and inducing compressive stress, and more specifically, the reinforcement layer **120** may include a bonding reinforcement layer and a compressive stress layer.

[0061] The bonding reinforcement layer is a layer introduced to increase bonding strength when the protective layer **140** is bonded onto the final support substrate **110** through the bonding layer **130** and is formed on the seed layer **150** or the final support substrate **110**, and it is preferable that a material constituting the bonding reinforcement layer be preferentially selected from SiO₂, SiN_x, etc.

[0062] The compressive stress layer is a layer that induces compressive stress and is formed on the bonding reinforcement layer, and the protective layer **140** or the second bonding layer B2 is formed on the compressive stress layer. The compressive stress layer is made of a dielectric material with a greater CTE than the CTE of the final support substrate **110**, for example, a material that relieves tensile stress, that is, induces compressive stress, such as AlN (4.6 ppm), AlNO (4.6 to 6.8 ppm), Al₂O₃ (6.8 ppm), SiC (4.8 ppm), SiCN (3.8 to 4.8 ppm), GaN (5.6 ppm), or GaNO (5.6 to 6.8 ppm), which serves to guide the quality improvement of products through stress control.

[0063] In the present invention, the above-described bonding reinforcement layer or compressive stress layer may be omitted in some cases, and in some cases, the entire reinforcement layer **120** may be omitted so that one surface of the seed layer **150** is in direct contact with the protective layer **140** or the final support substrate **110** is in direct contact with the second bonding layer B2. This case may be a structure that induces compressive stress together with a bonding function by depositing a material having a greater CTE than the final support substrate **110** on the protective

layer **140** or the bonding layer **130**.

[0064] Meanwhile, before depositing the reinforcement layer **120**, a surface planarization layer may be introduced on the support substrate **110**, and the surface planarization layer may be formed through a flowable oxide (FOX) such as SOG (liquid SiO₂) or HSQ.

[0065] The removing operation **S140** is an operation of removing the initial growth substrate **G1** to expose the metal polar surface of the seed layer **150**.

[0066] The removing operation **S140** includes performing mechanical polishing (grinding and lapping) on a back surface of the Si growth substrate **G1** and then removing the remaining thin Si growth substrate **G1** using a CLO process, and specifically, includes completely removing the remaining thin Si growth substrate **G1** through wet etching using a tetramethylammonium hydroxide (TMAH) or hydrofluoric+nitric+acetic acids (HNA) solution.

[0067] In this case, the removing operation **S140** includes etching and removing all of the SiN layer **P1**, the buffer layer **U1**, and the compressive stress introduction layer **R** that are formed in the growing operation **S110** as well as the growth substrate **G1** to expose the metal polar surface of the seed layer **150** transferred onto the final support substrate **110** and additionally etching and removing a region of the seed layer **150** on which Mg doping or carrier gas modulation is performed when Mg doping or carrier gas modulation is performed on the seed layer **150** to expose the metal polar surface of the seed layer **150** transferred onto the final support substrate **110**.

[0068] The group III nitride semiconductor template according to the present invention manufactured through the above-described process may basically have a structure in which the support substrate **110**, the bonding layer **130**, and the seed layer **150** are sequentially stacked, and the device active layer **160** made of a high-quality group III nitride semiconductor may be grown on the seed layer **150** transferred onto the final support substrate **110** and having the exposed metal polar surface.

[0069] Meanwhile, in some cases, the remaining thin Si growth substrate **G1** after mechanical polishing may be removed through a plasma dry process such as SF₆ to be completely removed, and heat treatment may be performed at a high temperature of 400° C. or higher to increase the bonding strength of the bonding layer **130** after mechanical polishing or after completely removing the remaining thin Si growth substrate **G1**.

[0070] FIG. **9** shows that the device active layer **160** is re-grown on the seed layer **150** of the group III nitride semiconductor template according to the method of manufacturing a group III nitride semiconductor template according to the first or second embodiment of the present invention.

[0071] After the group III nitride semiconductor template is manufactured through the above-described operations, the method according to the present invention may further include a re-growing operation, and as shown in FIG. **9**, the re-growing operation is an operation of re-growing the device active layer **160** on the seed layer **150** having the exposed metal polar surface.

[0072] When the device active layer **160** for a micro LED device for a display or a UV LED device having a vertical chip structure is re-grown, the material of the seed layer **150** of the group III nitride semiconductor template according to the present invention may include GaN, InGaN, or AlN, and the material of the final support substrate **110** may include optically transparent sapphire (using a laser lift off (LLO) process) or AlNcera (using a CLO process) having the same CTE as the group III nitride.

[0073] In addition, when the device active layer **160** for a high-frequency and switching power semiconductor device is re-grown, the material of the seed layer **150** of the group III nitride semiconductor template according to the present invention may include GaN or AlN, and the material of the final support substrate **110** may include AlNcera (using a CLO process) having the same CTE as the group III nitride, SiC having high heat dissipation performance and a small difference in CTE from the group III nitride, or Si that enables the manufacturing of large-diameter wafers and has high cost-effectiveness.

[0074] In addition, when the device active layer **160** for an AlN BAW or SAW filter device for 5G

or Wi-Fi communication is re-grown, the material of the seed layer **150** of the group III nitride semiconductor template according to the present invention may include AlN, and the material of the final support substrate **110** may include AlNcera (using a CLO process) having the same CTE as the group III nitride, SiC having high heat dissipation performance and having a small difference in CTE from the group III nitride, or optically transparent sapphire (using an LLO process). [0075] Hereinafter, a method **S200** of manufacturing the group III nitride semiconductor template according to the second embodiment of the present invention will be described in detail with reference to the accompanying drawings.

[0076] FIG. **5** is a flowchart of the method **S200** of manufacturing a group III nitride semiconductor template according to the second embodiment of the present invention, and FIG. **6** shows a process of manufacturing a group III nitride semiconductor template according to the second embodiment of the present invention.

[0077] As shown in FIGS. **5** and **6**, the method **S200** of manufacturing a group III nitride semiconductor template according to the second embodiment of the present invention includes a growing operation **S210**, a depositing operation **S220**, a bonding operation **S230**, and a removing operation **S240**.

[0078] The growing operation **S210** is an operation of growing a single-layer or multilayered group III nitride semiconductor seed layer **150** whose upper surface is a nitrogen polar surface on an initial growth substrate **G2**.

[0079] In the present embodiment, the initial growth substrate **G2** may be provided as a sapphire (α -phase Al₂O₃) substrate that may theoretically transmit 100% of a laser beam (single wavelength light) without absorption in the LLO process of the removing operation **S240** to be described below and which is optically transparent and has high-temperature heat resistance. In addition, the initial growth substrate **G2** may be provided as a patterned sapphire substrate (PSS) having a protrusion shape that is patterned regularly or irregularly in various dimensions (size and shape) in microscale or nanoscale to minimize crystal defects inside the group III nitride semiconductor thin film grown thereon.

[0080] FIGS. **7** and **8** show that the seed layer **150** whose upper surface is a nitrogen polar surface is grown through the growing operation **S210** of the method **S200** of manufacturing a group III nitride semiconductor template according to the second embodiment of the present invention.

[0081] More specifically, the growing operation **S210** includes preparing the initial growth substrate **G2** on which the CMP process is performed, performing heat treatment on the initial growth substrate **G2** at a high temperature of 1000° C. or higher in a hydrogen (H₂) atmosphere, performing nitriding treatment on a surface of the growth substrate **G2** at a high temperature of 1000° C. or higher in an ammonia atmosphere (NH₃ and N₂) to form an AlN layer **P2** having a film or island shape in a thickness of less than 5 nm, depositing the GaN or AlN buffer layer **U2** (or a nucleation layer) on the AlN layer **P2** in a thickness of 50 nm or less at a high temperature of 850° C. or higher, and growing the seed layer **150** on the buffer layer **U2** in a thickness of 2 μ m or less.

[0082] As shown in FIG. **7**, according to the growing operation **S210**, a structure in which the AlN layer **P2**, the GaN or AlN buffer layer **U2**, and the seed layer **150** are sequentially stacked is formed on the growth substrate **G2**, and the seed layer **150** having a nitrogen polar surface may be grown through the AlN layer **P2** and the buffer layer **U2** according to high-temperature nitriding treatment.

[0083] Meanwhile, as shown in FIG. **8**, to reinforce the nitrogen polarity of the surface of the seed layer **150**, Mg doping or carrier gas modulation may be performed on the seed layer **150** during growth.

[0084] Meanwhile, a channel through which gas generated during the process may move may be formed inside the seed layer **150** through an intentional patterning design.

[0085] Since the depositing operation **S220** and the bonding operation **S230** are the same as the

above-described method of manufacturing a group III nitride semiconductor template according to the first embodiment of the present invention, overlapping descriptions thereof will be omitted.

[0086] The removing operation **S240** is an operation of removing the initial growth substrate **G2** to expose the metal polar surface of the seed layer **150**.

[0087] The removing operation **S240** includes removing the sapphire growth substrate **G2** using an LLO process and specifically, includes irradiating a back surface of the optically transparent sapphire growth substrate **G2** with a laser beam to separate the initial growth substrate **G2** from the group III nitride semiconductor.

[0088] In this case, the removing operation **S240** includes etching and removing both the AlN layer **P2** and the buffer layer **U2** that are formed in the growing operation **S210** as well as the growth substrate **G2** to expose the metal polar surface of the seed layer **150** transferred onto the final support substrate **110** and additionally etching and removing a region of the seed layer **150** on which Mg doping or carrier gas modulation is performed when the Mg doing or carrier gas modulation is performed on the seed layer **150** to expose the metal polar surface of the seed layer **150** transferred onto the final support substrate **110**.

[0089] In this way, the device active layer **160** made of a high-quality group III nitride semiconductor may be re-grown on the seed layer **150** transferred onto the final support substrate **110** and having the exposed metal polar surface.

[0090] Meanwhile, in some cases, heat treatment may be performed at a high temperature of 400° C. or higher to increase the bonding strength of the bonding layer **130** after removing the sapphire growth substrate **G2**.

[0091] Since the re-growing operation is the same as the method of manufacturing a group III nitride semiconductor template according to the first embodiment of the present invention, overlapping descriptions thereof will be omitted.

[0092] As described above, although all the components constituting embodiments of the present invention are described as being combined or combined to operate as one, the present invention is not necessarily limited to these embodiments. That is, one or more of all the components may be combined to operate as one without departing from the scope of the purpose of the present invention.

[0093] In addition, the terms such as “comprise,” “constitute,” or “have” described above mean that the corresponding component can be included unless otherwise stated, and thus should be construed as further including another component rather than excluding another component. All terms including technical or scientific terms have the same meaning as commonly understood by those skilled in the art to which the present invention pertains unless defined otherwise. Commonly used terms, such as terms defined in dictionaries, should be interpreted as being consistent with the contextual meaning of the related art and are not interpreted in an ideal or excessively formal meaning unless explicitly defined herein.

[0094] In addition, the above description is merely an exemplary description of the technical spirit of the present invention, and those skilled in the art to which the present invention pertains will be able to variously modify and change the present invention without departing from the essential characteristics of the present invention.

[0095] Therefore, the embodiments disclosed in the present invention are not intended to limit the technical spirit of the present invention, but intended to describe the same, and the scope of the technical spirit of the present invention is not limited by these embodiments. The scope of the present invention should be construed by the appended claims, and all technical ideas within the equivalent scope should be construed as being included in the scope of the present invention.

Claims

- 1.** A method of manufacturing a group III nitride semiconductor template, the method comprising: a growing operation of growing a seed layer whose upper surface is a nitrogen polar surface on a growth substrate; a depositing operation of depositing a protective layer on the seed layer; a bonding operation of bonding the protective layer to a support substrate through a bonding layer; and a removing operation of removing the growth substrate to expose a metal polar surface of the seed layer.
 - 2.** The method of claim 1, wherein the growth substrate is a silicon (Si) substrate.
 - 3.** The method of claim 2, wherein the growing operation includes: performing high-temperature heat treatment on the growth substrate in a hydrogen atmosphere; performing high-temperature nitriding treatment on a surface of the growth substrate in an ammonia atmosphere to form an SiN layer; depositing a buffer layer on the SiN layer; growing a compressive stress introduction layer on the buffer layer; and growing a seed layer having a nitrogen polar surface on the compressive stress introduction layer.
 - 4.** The method of claim 3, wherein the removing operation includes removing the growth substrate, the SiN layer, the buffer layer, and the compressive stress introduction layer to expose the metal polar surface of the seed layer.
 - 5.** The method of claim 3, wherein Mg doping or carrier gas modulation is performed on at least one of the compressive stress introduction layer and the seed layer during growth.
 - 6.** The method of claim 5, wherein the removing operation includes removing the growth substrate, the SiN layer, the buffer layer, the compressive stress introduction layer, and a region of the seed layer on which the Mg doping or carrier gas modulation is performed to expose the metal polar surface of the seed layer.
 - 7.** The method of claim 1, wherein the growth substrate is a sapphire (Al.sub.2O.sub.3) substrate.
 - 8.** The method of claim 7, wherein the growing operation includes: a first operation of performing high-temperature heat treatment on the growth substrate in a hydrogen atmosphere; a second operation of performing high-temperature nitriding treatment on a surface of the growth substrate in an ammonia atmosphere to form an AlN layer; a third operation of depositing a buffer layer on the AlN layer; and a fourth operation of growing a seed layer having a nitrogen polar surface on the buffer layer.
 - 9.** The method of claim 8, wherein the removing operation includes removing the growth substrate, the AlN layer, and the buffer layer to expose the metal polar surface of the seed layer.
 - 10.** The method of claim 8, wherein Mg doping or carrier gas modulation is performed on the seed layer during growth.
 - 11.** The method of claim 10, wherein the removing operation includes removing the growth substrate, the AlN layer, the buffer layer, and a region of the seed layer on which the Mg doping or carrier gas modulation is performed to expose the metal polar surface of the seed layer.
 - 12.** The method of claim 1, wherein the depositing operation includes forming a reinforcement layer, which increases bonding strength and induces compressive stress, on the seed layer and then depositing the protective layer on the reinforcement layer.
 - 13.** The method of claim 1, wherein the bonding operation includes forming a reinforcement layer, which increases bonding strength and induces compressive stress, on the support substrate and then bonding the protective layer to the reinforcement layer through a bonding layer.
 - 14.** The method of claim 1, further comprising a re-growing operation of re-growing an device active layer on the exposed seed layer.
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